

Model Execution & Technical Data Specifications

Classification Model Operating Guide

1. Model Overview

Item	Details
Architecture	MobileNetV3, optimized and quantized (TorchScript / INT8)
Model file	static_int8_model.pt (or the weights file .pth)
Base training data	EuroSAT dataset, based on the Sentinel-2 satellite

2. Required Input Data Specifications

Parameter	Value used in training	Technical notes
Data source	Sentinel-2 L2A	Surface reflectance readings
Spectral bands	6 bands: B2 (Blue), B3 (Green), B4 (Red), B8 (NIR), B11 (SWIR1), B12 (SWIR2)	Band order in code: BANDS_SELECTED = [1, 2, 3, 7, 10, 11] (indices follow the Sentinel-2 band array)
Spatial dimensions	64 x 64 pixels	The code automatically resizes images of other sizes
Normalization	Divide by 10000.0	Converts raw reflectance values to the range 0 to 1

3. Inference Instructions

- Set the paths:** update PATH_MODEL to the location of the weights file, and path_test to the path of the TIF image to be tested.
- Input:** the input image should preferably be TIF / GeoTIFF and contain the spectral bands listed above.

4. Technical Notice: Other Satellites (Cross-Satellite Domain Shift)

If you want to test the model on imagery from other satellites (such as Landsat, WorldView, or proprietary sensors), it will **not** give accurate results directly without fine-tuning, for the following reasons:

- Spectral mismatch:** band wavelengths differ between satellites even when band names are similar.
- Scaling factor mismatch:** not all satellites use the 10000.0 divisor for reflectance.
- Spatial resolution mismatch:** a different pixel size affects the patterns the model has learned.

5. Fine-Tuning Requirements for a New Satellite

To apply the model to data from another satellite:

- 1 **Re-match the bands:** order the input bands by physical wavelength, not by channel number.
- 2 **Adjust normalization:** set the offset and divisor according to the value range of the new satellite.
- 3 **Fine-tune:** freeze the encoder layers and retrain the final classifier layers on a sample of the new satellite's data, using a small learning rate (e.g., 10^{-4}).

This document accompanies the inference code and the model weights file.